Unit 2

Language Syntax and Semantics

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- Dictionary based approach, Latent Semantic Analysis

Syntactic Analysis: Syntactic Representations of Natural Language

- Syntactic analysis, or parsing, focuses on **determining the grammatical structure** of sentences in natural language.
- It identifies **how words combine to form phrases, clauses**, and complete sentences, **adhering to the syntactic rules** of a given language.

Key syntactic representations include:

A) Constituency-Based Representations

- **Parse Trees**: Hierarchical structures where **nodes** represent **syntactic categories** (e.g., NP, VP), and **leaves** correspond to **words.**
- **Phrase Structure Grammar**: Frameworks like Context-Free Grammars (CFG) underlie these representations.

B) Dependency-Based Representations

- **Dependency Trees**: Represent relationships between words using directed edges, emphasizing head-dependent structures.
- **Dependency Parsing**: Efficient for languages with flexible word order.

C) Hybrid Representations

- Combine constituency and dependency representations for richer syntactic insights.

Parse Trees

- A **parse tree** (or **syntax tree**) is a tree structure that represents the syntactic structure of a sentence according to a given grammar, typically a **context-free grammar (CFG)**.

- It visualizes **how words** in a sentence are grouped into phrases and how these **phrases relate to one another hierarchically**.

- Parse trees are fundamental tools in computational linguistics and NLP, enabling deeper linguistic insights and effective language processing.

Components of a Parse Tree

1. **Root Node**: Represents the start symbol of the grammar, often denoted as S (Sentence).

2. **Non-Terminal Nodes**: Represent intermediate syntactic categories like noun phrases (NP), verb phrases (VP), or prepositional phrases (PP).

3. **Terminal Nodes**: Represent the actual words (or tokens) in the sentence.

Example

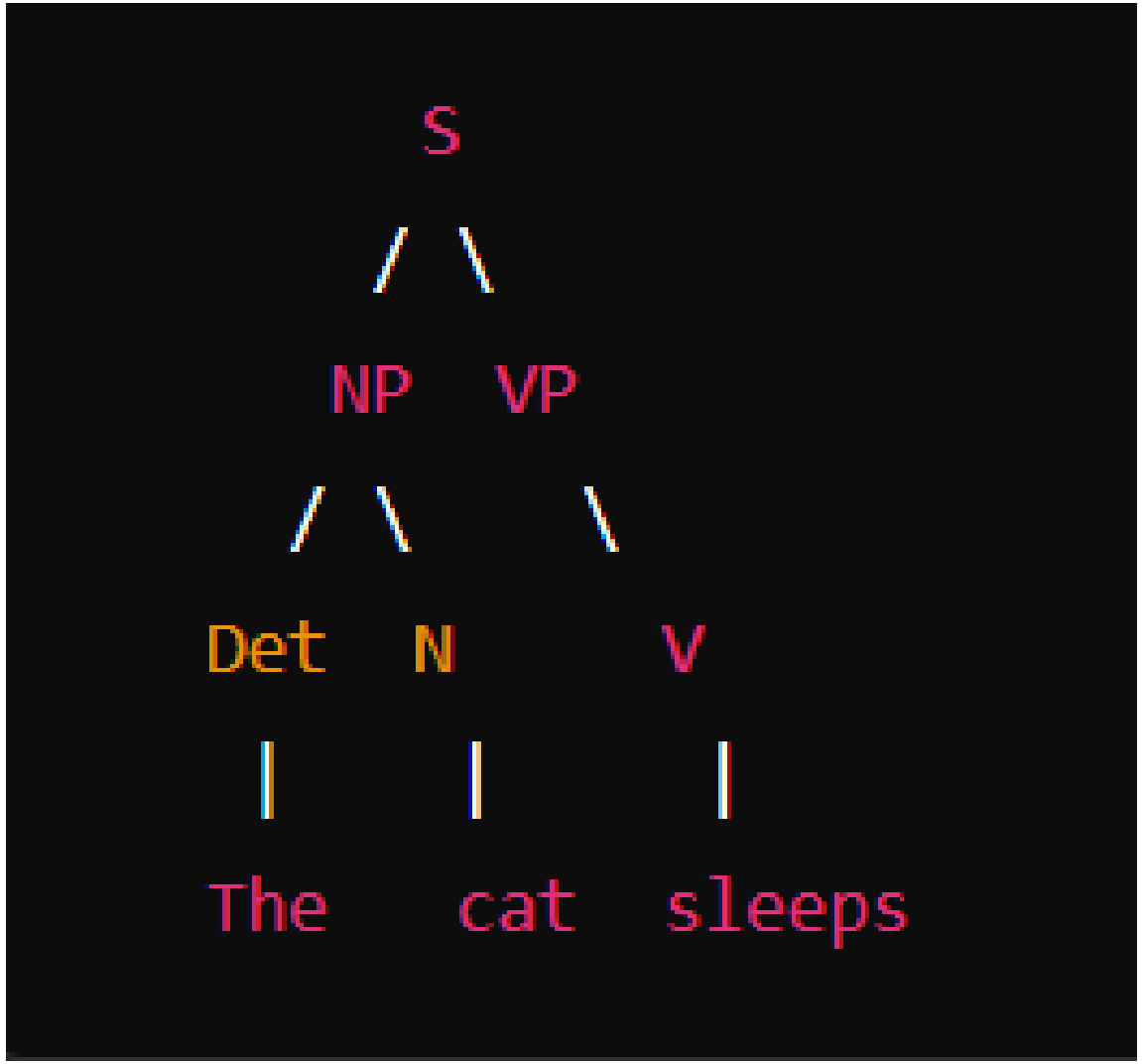
Consider the sentence:

"The cat sleeps.“

Using a simplified CFG:

1. S → NP VP
2. NP → Det N
3. VP → V
4. Det → "The"
5. N → "cat"
6. V → "sleeps"

The parse tree looks like this:

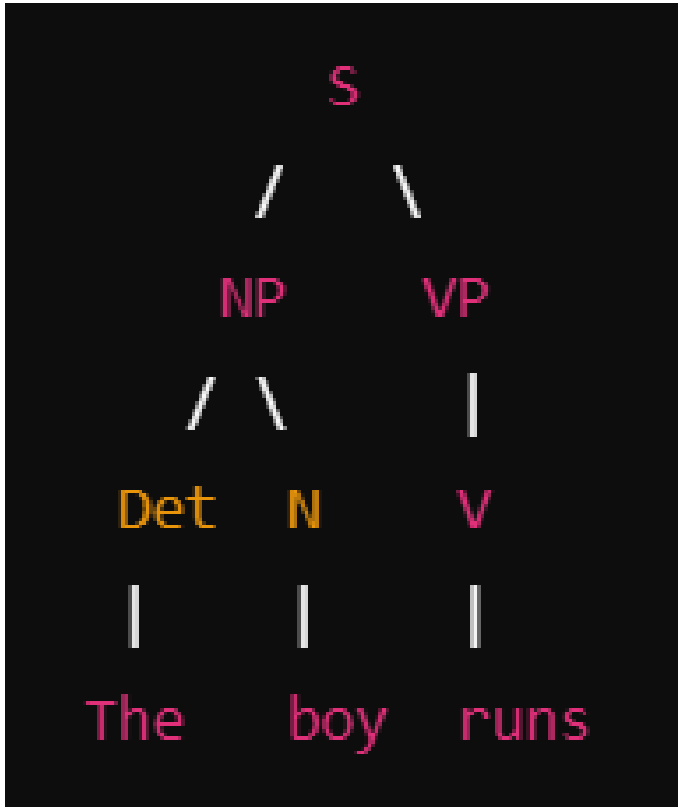


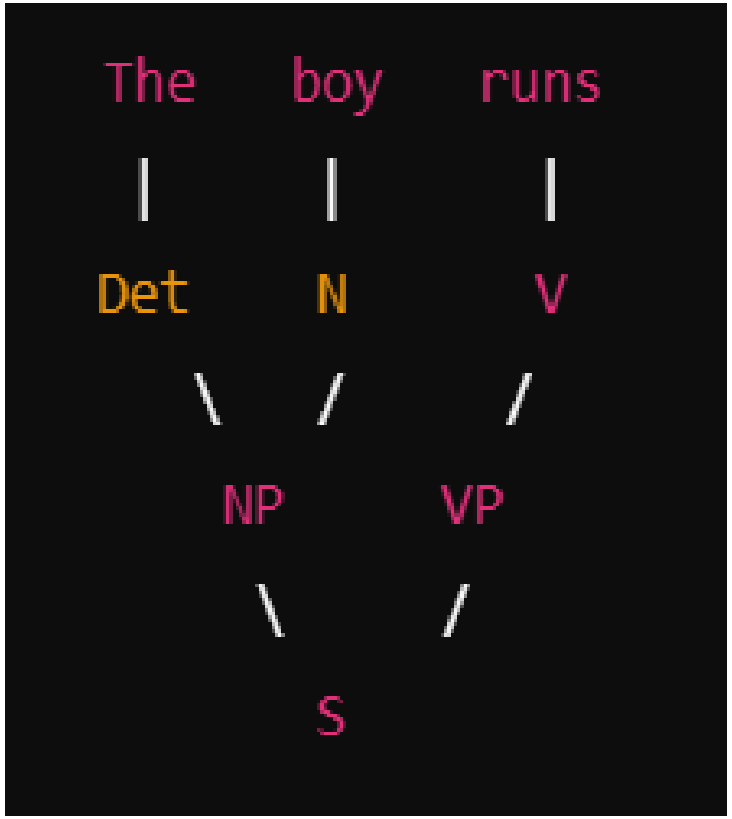
How Parse Trees Are Generated

- **Top-Down Parsing**: Start from the root (S) and expand nodes based on grammar rules until terminals match the input sentence.

- **Bottom-Up Parsing**: Start with the words (terminals) and combine them using grammar rules until the root (S) is derived.

Sentence: "The boy runs."





- Start with S → Expand into NP (Noun Phrase) and VP (Verb Phrase).
- Expand NP into Det (Determiner) and N (Noun).
- VP expands into V (Verb).

Begin with Det, N, and V → Combine Det and N into NP.

Combine NP and V into S.

Sentence: "She plays piano."

```
      S
    /   \
   NP    VP
    |    / \
    N   V   NP
        |   |
      plays piano
```

```
She   plays   piano
 |      |       |
 N      V       N
  \    /         \
   NP  VP         NP
       \          /
        S
```

Sentence: "Birds fly."

```
      S
    /   \
   NP    VP
    |     |
    N     V
    |     |
  Birds  fly
```

```
  Birds    fly
    |       |
    N       V
     \     /
      NP  VP
       \ /
        S
```

Sentence:
“ She Plays Piano”

"The dog chased the ball."

```
          S
        /   \
      NP     VP
     / \    /  \
   Det  N   V    NP
    |   |   |   / \
   The dog chased Det N
                   |  |
                  the ball
```

```
The   dog   chased   the   ball
 |     |      |      |     |
Det    N      V     Det    N
  \   /        \   /      |
   NP           VP        NP
    \          /         /
       S
```

“The dark brown fox was playing with white fat bear”

“The young curious child was reading a small colorful book.”

```
                S
              /
            NP
          /  |  \            \
        Det  Adj  Adj        VP
         |    |    |         /       \
        the  dark brown   Aux      VP
                   |       |          |
                  fox    was        V'
                                   /   \
                                 V     PP
                                 |     /       \
                              playing  P         NP
                                       |        /  |   \
                                     with     Adj  Adj   N
                                                |    |    |
                                             white fat  bear
```

```
        S
       /   \
     NP     VP
   / | \      \
 Det Adj Adj   VP
  |   |   |     / \
 the young curious Aux   VP
            |      |   / \
          child   was V   NP
                     |   / | \
                  reading Det Adj Adj N
                          |   |   |   |
                          a small colorful book
```

Phrase Structure Grammar

- Phrase Structure Grammar is a **type of formal grammar** used to represent the syntactic structure of sentences. Frameworks like **Context-Free Grammars (CFG) play a foundational role** in this representation.

- A set of rules that describe how words and phrases combine to form sentences in a language.

- It defines the hierarchical structure of a sentence by breaking it into smaller constituents or phrases (e.g., noun phrases, verb phrases).

Role of Context-Free Grammars (CFG) in NLP

1.Definition of CFG:

CFG is a type of grammar where **production rules** are used **to generate sentences.** Each rule has a single **non-terminal symbol (like S, NP, VP)** on the left-hand side and a **combination of terminal and non-terminal symbols on the right-hand** side.

$S \rightarrow NP\ VP$

$NP \rightarrow Det\ N$

$VP \rightarrow V\ NP$

2. Tree Representation:

CFG helps in constructing **parse trees**, which visually depict the syntactic structure of a sentence.

Applications in NLP:

- **Parsing**: CFGs are widely used **in parsers to analyze sentence structure**. Tools like **Stanford Parser** and **spaCy** implement such approaches.

- **Syntax Checking**: CFG **ensures that a sentence conforms** to the **grammatical rules** of a language.

- **Machine Translation**: By understanding sentence structure, CFG aids in **generating syntactically correct translations.**

- **Text Generation**: CFG helps in generating text that adheres to grammatical rules.

Limitations of CFG in NLP:

- CFGs assume that the structure of a language can be fully captured by context-free rules, but **natural languages often exhibit dependencies** (e.g., subject-verb agreement, long-distance dependencies) that are not easily handled by CFG alone.

- Extensions like **Lexicalized CFGs** or **Tree-Adjoining Grammars (TAGs)** are used to address these complexities.

Dependency-Based Representations

- Dependency-based representations **focus on the syntactic relationships** between words in a sentence. Instead of relying on **phrase structure (like in constituency-based** parsing), dependency representations emphasize the grammatical dependencies between individual words.

- **Key Features:**

1.Head-Dependent Structure:

1. Each word in a sentence (except the root) is dependent on another word, which acts as its **head**.
2. For example, in the sentence *"She eats an apple"*, the word ***"eats"*** **is the head**, and *"She"*, *"an"*, and *"apple"* are its dependents.

Directed Graph Representation:
Sentences are represented as directed graphs or trees, where:

- Nodes correspond to words.
- Edges represent dependencies (e.g., subject, object, modifier).

Grammatical Relations:
Dependencies are labeled with grammatical relations such as subject (*nsubj*), object (*obj*), and modifier (*amod*).

Non-Projectivity:
Some languages (e.g., German) exhibit non-projective dependencies, where dependencies can cross, making them more challenging to parse.

Dependency Trees

- A dependency tree is a tree structure that represents the syntactic structure of a sentence based on dependency relations.

Characteristics:

1.Single Root:

- The tree has a single root node, typically the main verb or predicate of the sentence.
- Example: In *"She quickly eats an apple"*, *"eats"* is the root.

2.Child Nodes:

- Words directly dependent on the root or other nodes are represented as child nodes.
- Each word (except the root) has exactly one parent.

3.Labeled Arcs:

- Arcs (edges) between words are labeled with dependency types (e.g., subject, object).

For the sentence *"She eats an apple"*, the dependency tree might look like this:

```
    eats (root)
   /   \
She   apple
        |
       an
```

- *She* → *eats*: Subject relation (*nsubj*).
- *eats* → *apple*: Object relation (*obj*).
- *apple* → *an*: Determiner relation (*det*).

Dependency Parsing

- Dependency parsing is the process of analyzing the grammatical structure of a sentence by **identifying dependency relationships between words** and constructing a dependency tree.

- **Types of Dependency Parsing:**

1.Projective Parsing:

- Assumes **no crossing dependencies**.
- Works well for languages with **simple structures** (e.g., English).

2.Non-Projective Parsing:

- Allows crossing dependencies.
- Necessary for languages like **Czech or German** with complex word order.

Parsing Methods:

1. **Rule-Based Parsers**:
   - Use hand-crafted rules to determine dependencies.
   - Example: Grammar-based parsers.
2. **Statistical Parsers**:
   - Use machine learning to predict dependencies based on training data.
   - Example: Transition-based parsers and graph-based parsers.
3. **Neural Parsers**:
   - Leverage deep learning models for dependency parsing.
   - Example: BERT-based parsers.

For the sentence: *"The quick brown fox jumps over the lazy dog"*, a dependency tree would look like:

```
          jumps (root)
        /      |      \
      fox     over     dog
    /  |       |        |
 The quick    the      lazy
          brown
```

Parse tree and dependency tree.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[1] 7s

```
!pip install spacy
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (fr
```

```
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 90.2 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy
nlp = spacy.load("en_core_web_sm")
print("spaCy loaded successfully!")
```

```
spaCy loaded successfully!
```

```python
import spacy
from spacy import displacy

nlp = spacy.load("en_core_web_sm")

text = "The quick brown fox jumps over the lazy dog"
doc = nlp(text)

for token in doc:
    print(token.text, token.dep_, token.head.text)

# Visualize dependency tree
displacy.render(doc, style="dep")
```

```
The det fox
quick amod fox
brown amod fox
fox nsubj jumps
jumps ROOT jumps
over prep jumps
the det dog
lazy amod dog
dog pobj over
```

Output Terms Explaination

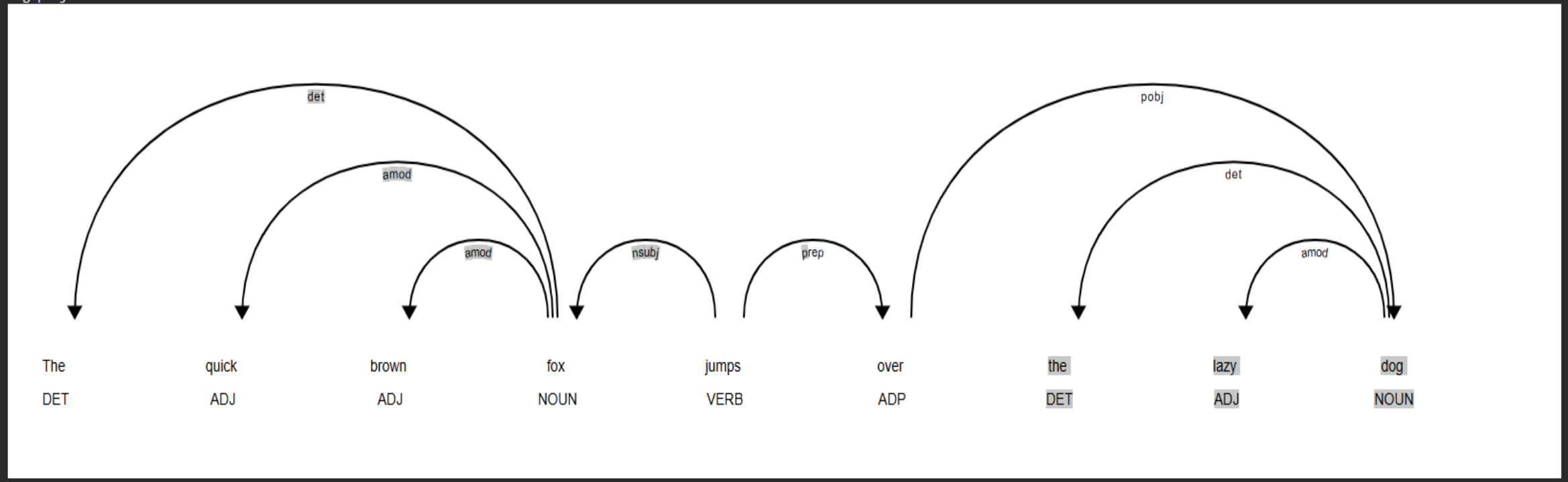
Token	Head	Dependency
The	fox	det
quick	fox	amod
brown	fox	amod
fox	jumps	nsubj
jumps	jumps	ROOT
over	jumps	prep
the	dog	det
lazy	dog	amod
dog	over	pobj

🔍 **Meaning:**

- **ROOT** → main verb of the sentence (`jumps`)
- **nsubj** → nominal subject (`fox`)
- **det** → determiner (`The`, `the`)
- **amod** → adjectival modifier (`quick`, `brown`, `lazy`)
- **prep** → preposition (`over`)
- **pobj** → object of preposition (`dog`)

```python
# Visualize dependency tree
displacy.render(doc, style="dep")
```

```
The det fox
quick amod fox
brown amod fox
fox nsubj jumps
jumps ROOT jumps
over prep jumps
the det dog
lazy amod dog
dog pobj over
```



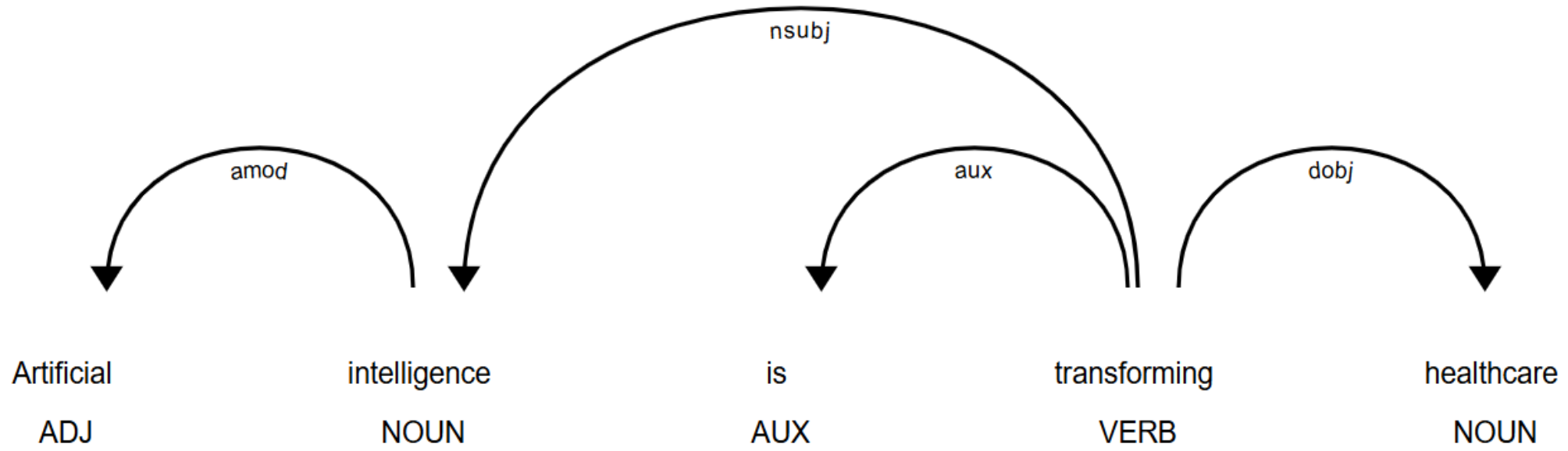
```python
sentences = [
    "She plays the piano",
    "Artificial intelligence is transforming healthcare"
]

for sent in sentences:
    doc = nlp(sent)
    print(f"\nSentence: {sent}")
    for token in doc:
        print(f"{token.text:15} {token.dep_:10} {token.head.text}")
  # Visualize dependency tree
displacy.render(doc, style="dep")
```

```
Sentence: She plays the piano
She             nsubj      plays
plays           ROOT       plays
the             det        piano
piano           dobj       plays

Sentence: Artificial intelligence is transforming healthcare
Artificial      amod       intelligence
intelligence    nsubj      transforming
is              aux        transforming
transforming    ROOT       transforming
healthcare      dobj       transforming
```

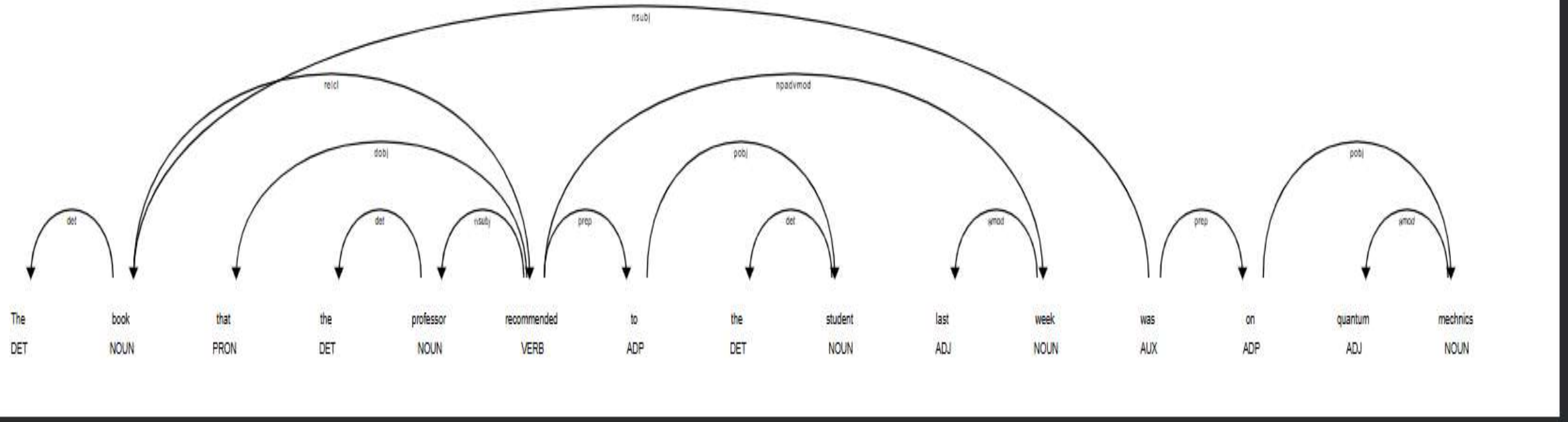
```
Sentence: Artificial intelligence is transforming healthcare
Artificial      amod       intelligence
intelligence    nsubj      transforming
is              aux        transforming
transforming    ROOT       transforming
healthcare      dobj       transforming
```



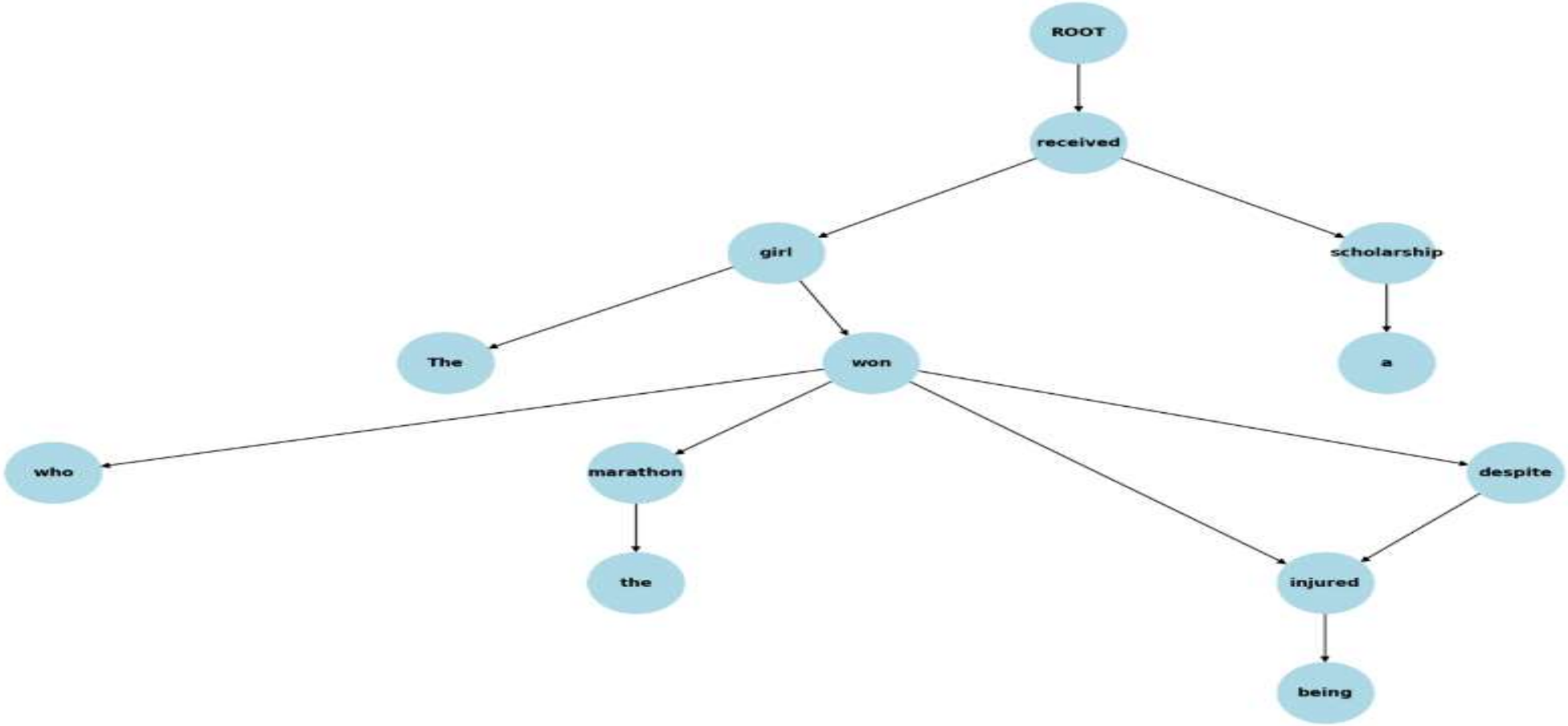
"The book that the professor recommended to the students last week was on quantum mechanics."

ROOT

```
Sentence: The book that the professor recommended to the student last week was on quantum mechnics
The          det       book
book         nsubj     was
that         dobj      recommended
the          det       professor
professor    nsubj     recommended
recommended  relcl     book
to           prep      recommended
the          det       student
student      pobj      to
last         amod      week
week         npadvmod  recommended
was          ROOT      was
on           prep      was
quantum      amod      mechnics
mechnics     pobj      on
```


nsubj
relcl
npadvmod
dobj
pobj
pobj
det
det
nsubj
prep
det
amod
prep
amod
The book that the professor recommended to the student last week was on quantum mechnics
DET NOUN PRON DET NOUN VERB ADP DET NOUN ADJ NOUN AUX ADP ADJ NOUN

"The girl who won the marathon despite being injured received a scholarship."



- **Common Libraries and Tools**

1. **SpaCy:** Lightweight, fast, and uses transition-based parsing.
2. **Stanza (StanfordNLP):** Neural pipeline supporting dependency parsing.
3. **UDPipe:** Pre-trained models for Universal Dependencies datasets.
4. **MaltParser:** Implements transition-based dependency parsing.
5. **ZPar:** A statistical dependency parsing library.

Projective vs. Non-Projective Dependency Trees

- Dependency trees can be classified as **projective** or **non-projective** based on whether the edges cross when drawn over the linear sequence of words in a sentence.

1. Projective Dependency Tree

In a projective tree:

- All edges can be drawn without crossing each other.
- This structure is typical for strict word-order languages like English.
- **Example Sentence (Projective):**
  "The cat chased the mouse."

Dependency Tree:

•chased → cat (subject)
•chased → mouse (object)
•cat → The (determiner)
•mouse → The (determiner)

```
  chased
   /   \
 cat   mouse
  |     |
 The   The
```

•The cat chased the mouse
•The mouse that the cat chased ran away. (Non Projective)

2. Non-Projective Dependency Tree

In a non-projective tree:

- At least one edge crosses another.
- This structure occurs in free-word-order languages like Hindi, Russian, or Czech.

Example Sentence (Non-Projective):

"The report, I told you yesterday, is on the table."

Dependency Tree:

- is → report (subject)
- is → on (prepositional phrase)
- on → table (object of preposition)
- report → told (relative clause)
- told → I (subject of told)
- told → yesterday (temporal modifier)
- told → you (object)

```
      is
     /  \
report  on
   |     \
  told   table
 / |  \
I you yesterday
```